

Chapter 3.

§ 3.1.

Definition of vector space and subspace.

two essential operations

i) vector addition

ii) scalar multiplication.

two conditions:

Let E be a set of vectors, E is a vector space/subspaces

iff

i) $V, W \in E \Rightarrow V + W \in E$

ii) $V \in E, \alpha \text{ a scalar} \Rightarrow \alpha V \in E$.

Remark: All linear combinations stay in the vector space.

Every vector space/subspace contains the zero vector.

Linear system $AX = b$.

Column space of A , denoted by $C(A)$, consists of all linear combinations of the columns of A .

$AX = b$ is solvable iff $b \in C(A)$.

If A is a $m \times n$ matrix, then $C(A)$ is a subspace of $\mathbb{R}^m$.

§ 3.2.

Defn: The null space of A denoted by $N(A)$, consists of all solutions to $AX = 0$. $N(A)$ is a subspace of $\mathbb{R}^n$.

— $N(A)$ consists of all linear combinations of special solutions of $AX = 0$. (solution process)

Defn: The $m \times n$ "staircase" U obtained by Gaussian elimination is called an echelon matrix. The reduced row echelon matrix R has zeros above the pivots as well as below and only 1's as the pivots.

§ 3.3.

Defn: The rank of a matrix A is the number of pivots and is often denoted by $r = \text{rank}(A)$.

i) $r \leq m$ and $r \leq n$.

ii) $r = m \rightarrow$ "full row rank", i.e. No zero rows in R .

$r = n \rightarrow$ "full column rank", i.e. No free variables.

Rank one matrix — every row is a multiple of the pivot row.

When $A \rightarrow R$.

i) $C(A) \neq C(R)$.

ii) $C(A^T) = C(R^T)$.

Null space matrix N has n rows and $n-r$ columns.

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} \begin{matrix} \uparrow r \text{ pivot rows} \\ \uparrow m-r \text{ zero rows} \end{matrix} \rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix} \begin{matrix} \uparrow r \text{ pivot variables} \\ \uparrow n-r \text{ free variables} \end{matrix}$$

$\downarrow$ $\downarrow$
 r pivot $n-r$
columns free variables

§ 3.4.

$AX = b$ $[A \ b] \rightarrow [R \ d]$

Complete solution: $X = X_p + X_n$

↳ The partial solution solves $AX_p = b$
 ↳ The $n-r$ special solution $\xrightarrow{R} RX_p = d$
 solve $AX_n = 0 \rightarrow RX_n = 0$

Full column rank ($r=n$) matrix A has

- i) All columns of A are pivot columns
- ii) no free variables or special solutions
- iii) $N(A) = \{0\}$
- iv) If $AX=b$ has a solution, then it has only one solution.

Full row rank ($r=m$) matrix A has

- i) All rows of A are pivot rows
- ii) $AX=b$ has a solution for any b .
- iii) $C(A) = \mathbb{R}^m$
- iv) There are $n-r=n-m$ special solutions.

§ 3.5

Defn: The columns of A are linearly independent when the only solution to $AX=0$ is $x=0$, i.e., $N(A) = \{0\}$.

- ii) The sequence of vectors $v_1, \dots, v_n$ are linearly independent if $c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$ only happens when $c_i = 0, i=1, 2, \dots, n$.

Defn: A set of vectors span a space if their linear combinations fill the space.

$C(A^T)$ — subspace of $\mathbb{R}^n$ spanned by rows of A .

$C(A)$ — subspace of $\mathbb{R}^m$ spanned by columns of A .

Defn: A basis for a vector space V is a sequence of vectors that has two properties:

- i) The vectors are linearly independent.
- ii) The vectors span the space V .

Defn: The dimension of a vector space V is the number of vectors in any of its basis, denoted by $\dim(V)$.

§3.6.

1. Row space of A , $C(A^T)$ is a subspace of $\mathbb{R}^n$ with dimension r
2. Column space of A , $C(A)$ is a subspace of $\mathbb{R}^m$ with dimension r
3. The Null space of A , $N(A)$ is a subspace of $\mathbb{R}^n$ with dimension $n-r$
4. The left null space of A , $N(A^T)$ is a subspace of $\mathbb{R}^m$ with dimension $m-r$.

Fundamental Theorem of Linear Algebra I

Chapter 4.

§4.1

Defn: Two vector spaces V and W are orthogonal if every vector v in V is perpendicular to every vector w in W , i.e.
 $v \cdot w = 0$ or $v^T w = 0$ for any $v \in V, w \in W$.

Defn: The orthogonal complement of V contains every vector that is perpendicular to V , is denoted by $V^\perp$.

$$N(A) = (C(A^T))^\perp$$

$$N(A^T) = (C(A))^\perp$$

Fundamental Theorem of Linear Algebra II

$$x \in \mathbb{R}^n, \quad x = x_r + x_n$$

$\hookrightarrow$ row space component $\hookrightarrow$ a null space component

Any n linearly independent vectors in $\mathbb{R}^n$ must span $\mathbb{R}^n$.

§4.2.

i) Projection of b onto the line through a

$$P = \left(\frac{a^T b}{a^T a} \right) a$$

Projection matrix: $P = \frac{aa^T}{a^T a}$

ii) Projection of b onto the space spanned by $a_1, \dots, a_n$

$$A = [a_1, a_2, \dots, a_n]$$

$$P = A(A^T A)^{-1} A^T b$$

Projection matrix: $P = A(A^T A)^{-1} A^T$

§4.3.

Consider $Ax = b$

Least squares solution $\hat{x}$

$$A^T A \hat{x} = A^T b$$

→ make $E = \|Ax - b\|$ as small as possible

Fitting by a straight line.

At time $t_1, \dots, t_m$ ($m \geq 2$)

heights $b_1, \dots, b_m$.

best line: $y = C + Dt$.

$$A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}, \quad X = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$A^T A \hat{x} = A^T b \rightarrow \hat{x}$$

Fitting by a parabola $y = C + Dt + Et^2$.

$$A = \begin{bmatrix} | & z_1 & z_1^2 \\ | & \vdots & \vdots \\ | & z_m & z_m^2 \end{bmatrix}, \quad X = \begin{bmatrix} C \\ D \\ E \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$A^T A \hat{X} = A^T b \rightarrow \hat{X}$$

§4.4.

Defn: The vectors $q_1, \dots, q_n$ are orthonormal if

$$q_i^T q_j = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

A matrix with orthonormal columns is assigned the special letter Q .

$$Q^T Q = I \Rightarrow Q^T = Q^{-1}$$

$$\begin{cases} \|QX\| = \|X\| \\ (QX)^T (QY) = X^T Y \end{cases}$$

The Gram-Schmidt Process

$$A \rightarrow Q$$

$$A = [a_1, a_2, \dots, a_n] \rightarrow Q = [q_1, q_2, \dots, q_n]$$

Algorithm:

$$\left\{ \begin{aligned} \tilde{q}_1 &= a_1 \\ \tilde{q}_2 &= a_2 - \frac{\tilde{q}_1^T a_2}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 \\ \tilde{q}_3 &= a_3 - \frac{\tilde{q}_1^T a_3}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 - \frac{\tilde{q}_2^T a_3}{\tilde{q}_2^T \tilde{q}_2} \tilde{q}_2 \\ &\vdots \\ \tilde{q}_n &= a_n - \frac{\tilde{q}_1^T a_n}{\tilde{q}_1^T \tilde{q}_1} \tilde{q}_1 - \frac{\tilde{q}_2^T a_n}{\tilde{q}_2^T \tilde{q}_2} \tilde{q}_2 - \dots - \frac{\tilde{q}_{n-1}^T a_n}{\tilde{q}_{n-1}^T \tilde{q}_{n-1}} \tilde{q}_{n-1} \end{aligned} \right\} \text{orthogonalization}$$

$$q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|}, \quad q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|}, \quad \dots, \quad q_n = \frac{\tilde{q}_n}{\|\tilde{q}_n\|} \quad \left. \vphantom{\frac{\tilde{q}_1}{\|\tilde{q}_1\|}} \right\} \text{normalization.}$$

QR factorization:

$$\underbrace{A = QR}_{\substack{Q \text{ orthonormal} \\ R \text{ upper triangular}}} \rightarrow \underline{R = Q^T A.}$$